INTERNSHIP MID-REVIEW PRESENTATION

EconoCausal

AI-Integrated Causal Demand Estimation & Dynamic Pricing Decision-Support System

A semiparametric econometric framework leveraging Double Machine Learning (DML) to isolate structural price elasticity, evaluate retail scenarios under inventory constraints, and enforce business governance.

CANDIDATE DETAILS

Student: Ippili Hasika
Role: Data Analyst Intern

ACADEMIC AFFILIATION

College: ANITS
Department: Information Technology

INTERNSHIP HOST

Organization: Infotact Solutions
Stage: Mid-Review Defense (Days 1–19)

CONTEXT & MOTIVATION

Problem Statement: The Retail Pricing Challenge

Why standard predictive machine learning leads to flawed pricing decisions and revenue destruction.

CORE BUSINESS CHALLENGES

- **Price vs. Demand Trade-off:**

  Price changes directly drive consumer purchasing. Raising prices may reduce volume sold.

- **The Discounting Fallacy:**

  Discounts may boost volume, but under inelastic demand, total revenue (P × Q) drops significantly.

- **Endogeneity & Confounding:**

  Store location, seasonality, inventory, and promotions jointly influence price and demand.

- **Predictive vs. Causal Gap:**

  Standard supervised models forecast demand under existing policies, but cannot evaluate counterfactuals.

- **Need for Governance:**

  Retailers require pricing engines bounded by inventory limits, cost realities, and uncertainty.

PROJECT SCOPE

Project Objectives: Bridging Prediction and Causality

Six structured objectives designed to transition from historical observation to decision intelligence.

1. Data Engineering & Auditing

Analyze 73,100 store-level retail records across 20 products, 5 stores, and 4 regions over 2 full years (2022–2024). Investigate seasonality, trends, and data quality issues to establish a reliable foundation for modeling.

2. Exploratory & Temporal Analysis

Develop time-series visualizations and statistical tests to uncover demand patterns, including weekly cycles, holiday effects, and long-term trends. Quantify how external factors like weather and pricing correlate with sales volume.

3. Baseline Demand Modeling

Build and benchmark predictive models (Ridge CV) to forecast daily sales. Use chronological hold-outs and time-series cross-validation to prevent data leakage and ensure realistic performance evaluation.

4. Double Machine Learning (DML)

Implement Robinson's partially linear semiparametric model with 3-fold cross-fitting and HC3 standard errors to isolate the causal effect of price on demand, controlling for confounding variables.

5. What-If Scenario Simulation

Develop a counterfactual engine to evaluate hypothetical pricing strategies (±10% adjustments), quantifying their impact on revenue and profit, and identifying optimal price points.

6. Decision-Support & Governance

Translate findings into actionable business insights, including elasticity-driven recommendations. Deliver an interactive Streamlit dashboard with 95% confidence intervals and model governance documentation.

DATA ARCHITECTURE

Dataset Description & Verified Quality Audit

Daily store-level inventory and sales transactions across two consecutive calendar years (73,100 records).

CORE TRANSACTION SCHEMA & DESCRIPTIVE STATISTICS

- **Date (731 days):**
  2022-01-01 to 2024-01-01. Transformed to Year, Month, DayOfWeek.
- **Product ID (20 items):**
  P0001 to P0020 across 5 distinct merchandise categories.
- **Store ID (5 stores):**
  S001 to S005 spanning 4 regions (East, North, South, West).
- **Category (5 departments):**
  Clothing, Electronics, Furniture, Groceries, Toys (~14.6k records each).
- **Price (Treatment T):**
  Mean: 55.03|*Median* : 55.01 | Range: [10.04, 99.97].
- **Units Sold (Outcome Y):**
  Mean: 140.2 units | Median: 107.0 units | Range: [0, 499 units].
- **Inventory Level:**
  Daily on-hand stock. Mean: 273.2 units | Range: [50, 500 units].
- **Gross Revenue (Derived):**
  Calculated as Price × Units Sold. Total Sales = $564.1 Million across 2 years.

VERIFIED QUALITY AUDIT

- **Total Rows:**
  73,100 verified records across 100 product-store pairs.
- **Missing Values:**
  0 nulls across all columns (100% complete dataset).
- **Duplicate Records:**
  0 duplicate rows detected in automated checks.
- **Intra-SKU Variation:**
  100% of product-store pairs exhibit temporal price changes.
- **Forecast Anomaly:**
  Identified negative Demand Forecasts (-9.99); isolated from causal model.
- **Missing Unit Cost:**
  Wholesale procurement cost (COGS) is absent. Mandates revenue optimization.

DATA ENGINEERING & EDA

Data Preprocessing & Exploratory Data Analysis

Systematic data sanitization, feature engineering, and empirical distributional diagnostics.

1. Data Sanitization

ISO datetime parsing, non-negative checks, 0 nulls, 0 duplicate records.

2. Feature Engineering

Engineered Year, Month, DayOfWeek, and Derived Revenue (Price × Units Sold).

3. Bivariate Finding

Unadjusted Price vs. Demand correlation r ≈ 0.00; OLS slope = +0.0045 (p=0.77).

METHODOLOGICAL PIPELINE

End-to-End Project Methodology and Workflow

A structured 10-stage architecture bridging data engineering, predictive modeling, DML, and pricing governance.

PREDICTIVE BENCHMARKING

Baseline Demand Modeling: Predictive vs. Causal

Evaluating supervised forecasting performance and uncovering the post-treatment inventory trap.

PREDICTIVE MODELING AUDIT

- **Chronological Split:**

  80% Train (58,480 rows) | 20% Test (14,620 rows). Avoids data leakage.

- **Supervised Models Evaluated:**
  - Random Forest: MAE = 69.51, R^2 = 0.3184
  - HistGradientBoost: MAE = 68.68, R^2 = 0.3419
  - RidgeCV (L2): MAE = 68.65, R^2 = 0.3443
  - Dummy Baseline: MAE = 88.81, R^2 = 0.00

- **Predictive vs Causal Distinction:**

  ML predicts E[Y | Price, Inventory] under past policy, but cannot evaluate causal interventions (∂Y/∂Price).

- **The Inventory Collider Trap:**

  Inventory Level drives 0.7118 test R^2 drop in permutation importance; Price accounts for only 0.0029! Inventory is a stocking reflection, not demand.

ECONOMETRIC FOUNDATION

Double Machine Learning (DML): Causal Identification

Robinson's partially linear model with Neyman orthogonality and 3-fold cross-fitting.

SCENARIO SIMULATION

Pricing Scenario Analysis: Constrained What-If Simulation

Evaluating candidate price changes (-10% to +10%) under physical inventory caps and uncertainty.

SIMULATION ENGINE SPECIFICATION

- **Demand Formula:**

  Q_hat(P) = min(InventoryCap, max(0, Q_0 + θ·ΔP)). Evaluates discrete price steps: -10%, -5%, Base, +5%, +10%.

- **Benchmark Product (P0001 at S001):**

  Base Price: 55.13|*BaseDailySales* : 173.0*units*|*BaseRevenue* : 9,537.49 | Stock Cap: 195 units.

- **Discounting Disaster (-10%):**

  Candidate Price = $49.62. Demand = 172.94 units. Realized Revenue = $8,580.81 (immediate -$956.68 or -10.03% collapse!).

- **Price Increase (+10%):**

  Candidate Price = 60.64. *Demand* = 173.06*units*. *SimulatedRevenue* = 10,494.82 (+10.04%).

- **Revenue ≠ Profit Warning:**

  Revenue is P × Q. Operating Profit is (P - c) × Q - F. Without wholesale cost (c), claiming +10% is optimal is invalid. Governance mandates micro-pilots (±2%).

EMPIRICAL FINDINGS

Results & Key Learnings: 18-Specification Robustness

Verified econometric parameters, confidence bounds, and multi-model stress testing.

VERIFIED PROJECT RESULTS

θ = +0.0107 units per $1 price change

95% CI: [-0.0197, +0.0411]

- **Primary DML Estimate (Spec 1):**
- HC3 Std Error: 0.0155 | p-value: 0.4899
- Structurally price-inelastic (fails to reject θ=0).

- **Collider Sensitivity (Spec 2):**

Adding Inventory Level yields θ = -0.0195 (p=0.119). Confirms severe collider shift when conditioning on post-treatment stock.

- **18/18 Specifications Span Zero:**

Zero out of 18 model specifications (different nuisance models, categories, regions, years) achieve significance at α=0.05.

- **Category Elasticities:**

Clothing (θ = -0.022), Electronics (θ = -0.0005), Furniture (θ = -0.015), Groceries (θ = +0.043), Toys (θ = +0.002). All span zero.

- **Strategic Business Rule:**

Strictly reject automated discounting. Demand elasticity does not justify price cuts.

CRITICAL APPRAISAL

Project Challenges, Limitations & Future Scope

Honest evaluation of data constraints, econometric trade-offs, and future research roadmap.

CHALLENGES & DATA LIMITATIONS

• Unobserved Confounding:

Observational sales data lacks marketing campaigns, promotional discounts, foot traffic, and competitor actions.

• Missing Unit Cost Data (COGS):

The database contains no wholesale cost information. Maximizing gross revenue is not equivalent to maximizing net profit.

• Inventory Timestamp Ambiguity:

If inventory is logged at end-of-day, conditioning on it introduces post-treatment collider bias.

• Linear Treatment Assumption:

Partially linear DML assumes constant marginal price sensitivity across candidate prices.

• Statistical Non-Significance:

With p=0.4899, pricing outputs serve as decision-support heuristics rather than absolute causal truths.

FUTURE RESEARCH & EXTENSIONS

• Heterogeneous Treatment Effects (HTE):

Deploy Causal Forests at generalized DML to explore whether specific luxury or perishable SKUs exhibit price sensitivity.

• Cost Modeling & Profit Optimization:

Acquire wholesale cost structures to enable true constrained profit maximization: Π = (P - c)·Q - F.

• Completed Streamlit Dashboard:

Productionized dashboard in app.py with 8 interactive modules, real-time what-if sliders, and governance matrices already built!

• A/B Micro-Pilot Testing:

Design a randomized store-level pricing pilot (±2% price changes) across select retail clusters to establish ground-truth causal elasticity.

• Cross-SKU Cannibalization:

Model cross-price elasticities to capture substitution effects across competing products within the same merchandise department.

MID-REVIEW PROJECT SUMMARY

Conclusion: EconoCausal Impact & Deliverables

- **Problem Solved:**

  Addressed the flaw of using naive predictive regression for retail pricing, which mistakes correlation for causation and overlooks inventory colliders.

- **Rigorous Methodology:**

  Integrated Double Machine Learning (Robinson 1988) with 3-fold cross-fitting, ExtraTrees nuisance models, and HC3 robust standard errors.

- **Verified Finding:**

  Demonstrated that demand is structurally inelastic ($\theta = +0.0107$, $p = 0.49$), proving that discounting destroys revenue across all 18 specifications.

- **Delivered Assets:**

  19 engineering daily logs, 13 verified Jupyter notebooks, comprehensive validation modules, and a fully interactive Streamlit application.

Thank You — Questions & Answers

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